

Omar Ads

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Robotics engineer focused on **reinforcement learning** and **behavior / controls** for legged and humanoid systems. Backed by 1.5 years of production AI/ML engineering, now applied to RL policy training (Isaac Lab, MuJoCo), sim-to-real, and ROS2 / C++ behavior systems.

Education

Purdue University

West Lafayette, IN, USA

M.Sc. Interdisciplinary Robotics

Jan 2026 – Present

- Coursework: **Optimization Methods for Systems and Control, Intro to Robotics, Multi-Agent Systems**
- Focus: **robot modeling, optimal control, and reinforcement learning** for manipulation and locomotion

Monash University

Selangor, Malaysia

B.Eng. Electrical and Computer Systems Engineering, CGPA: 3.6

Feb 2020 – Dec 2023

- Minor in **Artificial Intelligence in Engineering** • Member of **Shell Eco-Marathon** Coding Team

Robotics & Reinforcement Learning Projects

Bimanual UR10 Handoff — GPU-Parallel RL

2026

Isaac Lab (DirectRLenv), RSL-RL, PPO, PyTorch, USD

github.com/OmarAds2002/bimanual-rl

- Trained two UR10 arms for a coordinated object handoff with **PPO** across **1024 parallel environments** on a single GPU: **84% success, zero drops, no curriculum**.
- Diagnosed a **weld-geometry collision** causing persistent policy plateaus, and resolved a hold-and-wait reward exploit with a **mutual-grasp constraint** at the transfer.

MuJoCo Pick-and-Place — PPO & SAC

2026

MuJoCo, Stable-Baselines3, PPO, SAC, Weights & Biases

github.com/OmarAds2002/mujoco-pick-and-place

- Built a 6-DOF torque-controlled environment; **SAC reached a 1.0 success rate** under domain randomization, with potential-based reward shaping (Ng 1999) and a reverse curriculum.
- Identified and fixed **four distinct reward exploits** (drag-farming, grip-cycling, ballistic fling, velocity-gate loop).

TurtleBot3 Autonomous Patrol — Behavior Trees + Nav2

2026

ROS2, BT.CPP v4, Nav2, Gazebo, C++

github.com/OmarAds2002/turtlebot3-autonomous-patrol

- Authored **7 custom C++ BT.CPP v4 nodes** over a full Nav2 stack; reactive object detection **interrupts navigation mid-drive** via an Inverter + ReactiveSequence pattern.
- Designed a standalone **patrol_commander** process owning the BT factory, ticking at **10 Hz** on a MultiThreadedExecutor; SLAM (slam_toolbox), AMCL, Gazebo Harmonic world.

Experience

AI/ML Engineer

Giza, Egypt

Dar El Handasah

Apr 2024 – Present

- Engineered scalable **LLM and RAG pipelines** (LangChain, OpenAI API, Azure AI) integrated into production **REST APIs**, reducing manual research time by ~40%.
- Built and deployed **containerized microservices** (Docker, CI/CD via GitHub Actions) for real-time data processing and multi-agent coordination modules.
- Automated data ingestion workflows in **Python (Pandas, SQLAlchemy)** and MongoDB, enabling downstream ML model training at scale.
- Shipped **full-stack AI applications** (React, FastAPI, PostgreSQL) serving 50+ internal engineering users.

Selected AI/ML Project

RAG-Powered Engineering Knowledge Base

2024 – 2025

Python, FastAPI, LangChain, FAISS, PostgreSQL, React

- Built an end-to-end **retrieval-augmented generation** system (document ingestion, FAISS vector search, FastAPI backend, React frontend), cutting document lookup time by **60%**.

Publications

Evaluation of Six Local Climate Zones Mapping Methods for UHI Studies

2026

Transactions in GIS (Wiley, Open Access)

- Co-authored research on **GeoAI and urban sustainability**; implemented **CNN-based classifiers** on Landsat 8 imagery achieving **~95% accuracy**, benchmarked against WUDAPT, GEE, GIS, and hybrid pipelines.

Skills

- **Robotics & RL:** Isaac Lab, Isaac Sim, MuJoCo, ROS2 (Python/C++), Nav2, BT.CPP v4, Gazebo, RL (PPO, SAC, DDPG), RSL-RL, Stable-Baselines3, domain randomization, curriculum learning, sim-to-sim / sim-to-real, reward shaping
- **Programming:** Python, C++, JavaScript/TypeScript, SQL
- **AI/ML:** PyTorch, TensorFlow, CNNs, Transformers, LLMs, RAG, LangChain, Weights & Biases, MLflow
- **Systems & Cloud:** Docker, Git, GitHub Actions (CI/CD), Linux, FastAPI, REST APIs, PostgreSQL, MongoDB, Azure, AWS
- **Languages:** English (Fluent), Arabic (Native)

Certifications

- Azure AI Engineer (AI-102), AI Fundamentals (AI-900), Fundamentals (AZ-900) • AWS AI Practitioner, Cloud Practitioner • Datacamp Associate Data Scientist